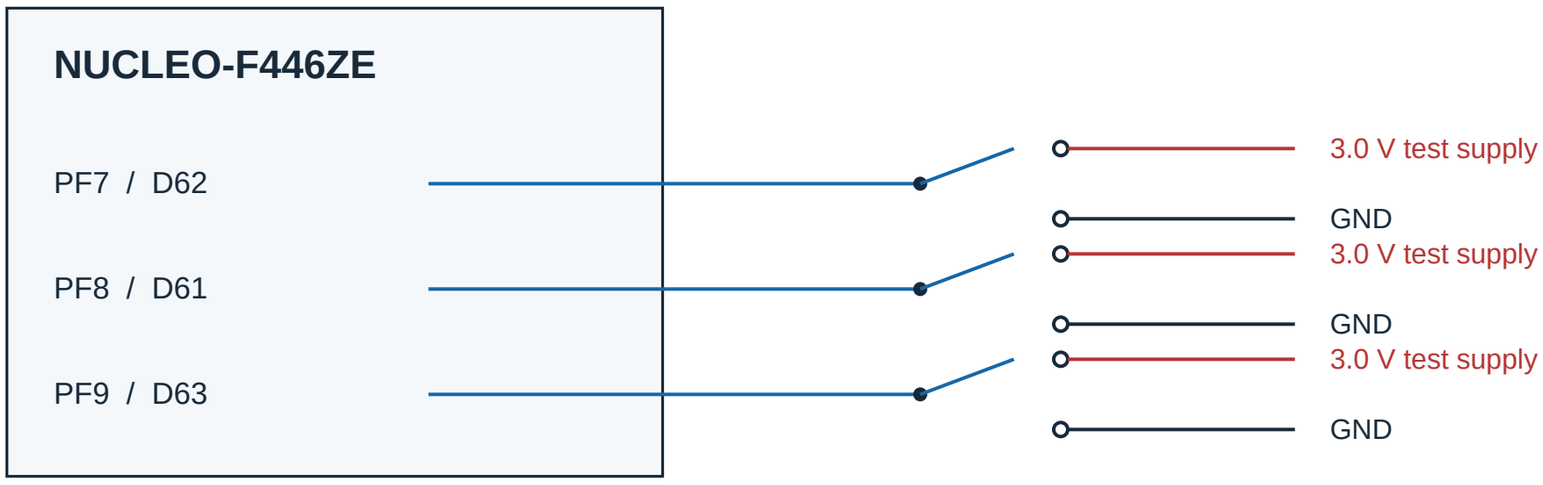
Part 1 - ADC + DMA test wiring

Use the female Arduino / Zio sockets. D-numbers below are board labels; PF7 etc. are MCU pin names.

Exercise	ADC input	Female socket	Header position
1.1 and 1.2	PF7 / ADC3 channel 5	D62	CN9 pin 26
1.2 only	PF8 / ADC3 channel 6	D61	CN9 pin 24
1.2 only	PF9 / ADC3 channel 7	D63	CN9 pin 28
Both	Ground	GND	CN8 pin 11 or pin 13



For each input:

move ONE jumper between 3.0 V and GND.

Do not join the two supply rails together.

USB powers the Nucleo and carries serial output. Use the ST-LINK USB connector.

Connect the test-supply negative terminal to Nucleo GND. Keep every ADC input at 0-3.3 V.

Part 1.1: connect PF7 only. Part 1.2: connect all three inputs; do not leave them floating.

Serial monitor: 115200 baud. Expect approximately 0 at GND and 3723 at 3.0 V (3.3 V reference).

The switch symbols represent jumper selection; physical switches are not required.

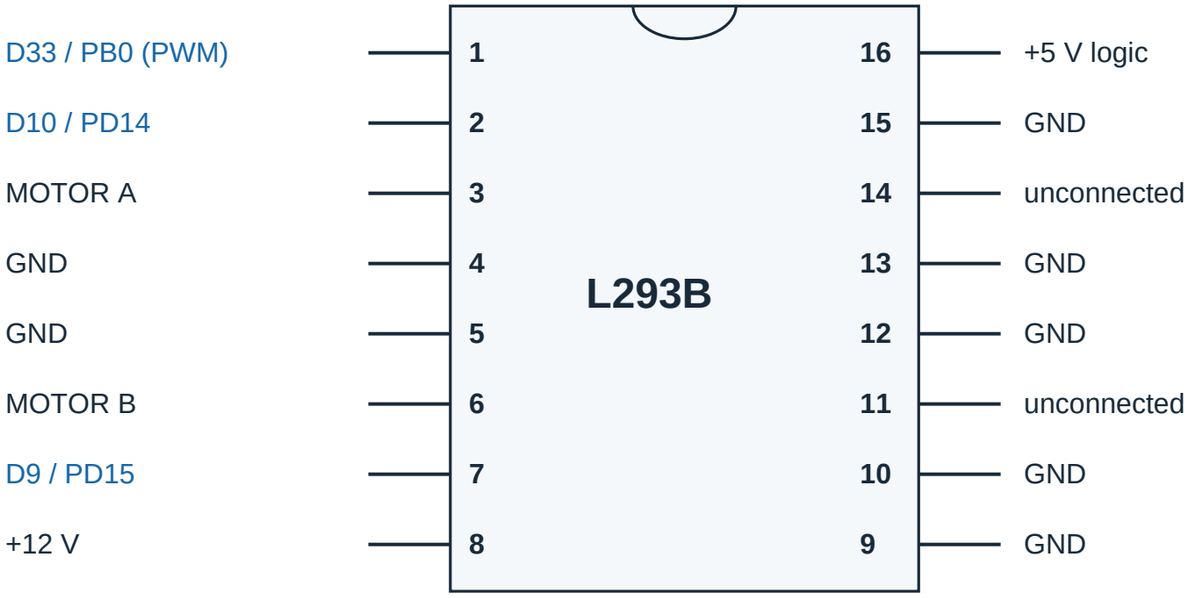
Pin source: ST UM1974, Table 18, pp. 57-59

Part 2 - Potentiometer + motor driver

Power the Nucleo through ST-LINK USB. Use an external regulated +5 V logic supply and +12 V motor supply.

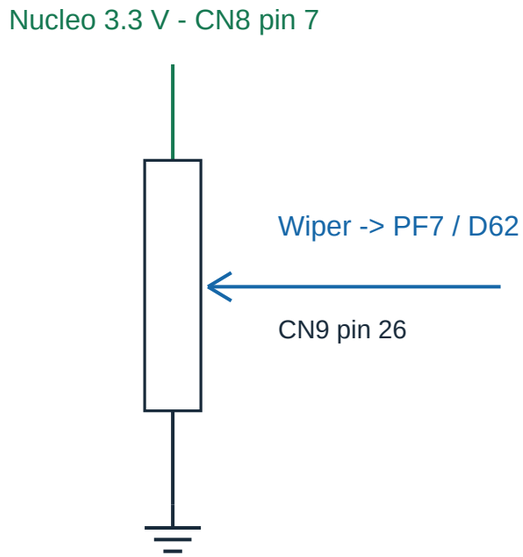
L293B, DIP-16 - TOP VIEW

Notch at the top. Equal net labels mean an electrical connection.



Pins 9, 10, 15: GND. Pins 11 and 14: leave open (unused bridge).

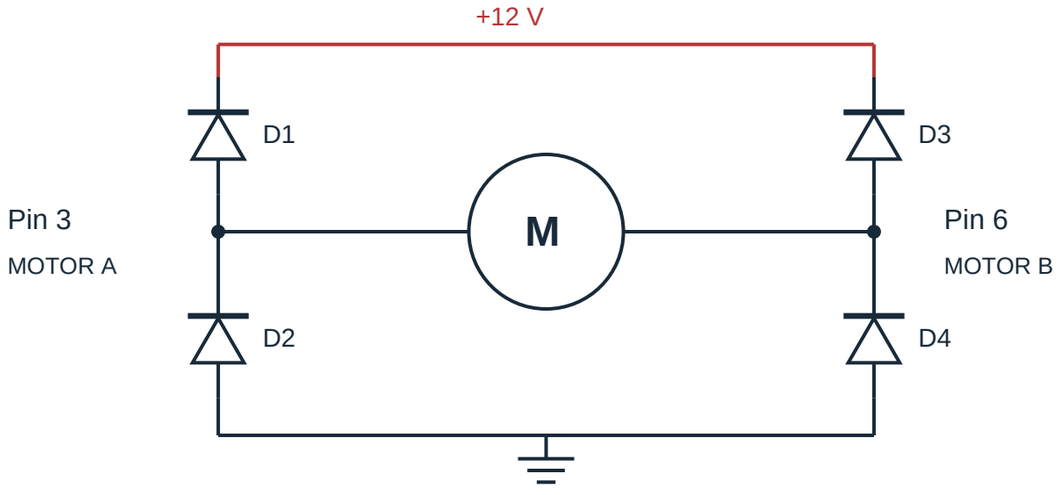
Potentiometer (10 kohm linear)



Outer terminals: 3.3 V and GND.

Wiper: ADC input. No 5 V connection here.

Four clamp diodes (1N4001 specified by the lab)



Diode band = cathode (K)

D1: anode -> pin 3; band -> +12 V

D2: anode -> GND; band -> pin 3

D3: anode -> pin 6; band -> +12 V

D4: anode -> GND; band -> pin 6

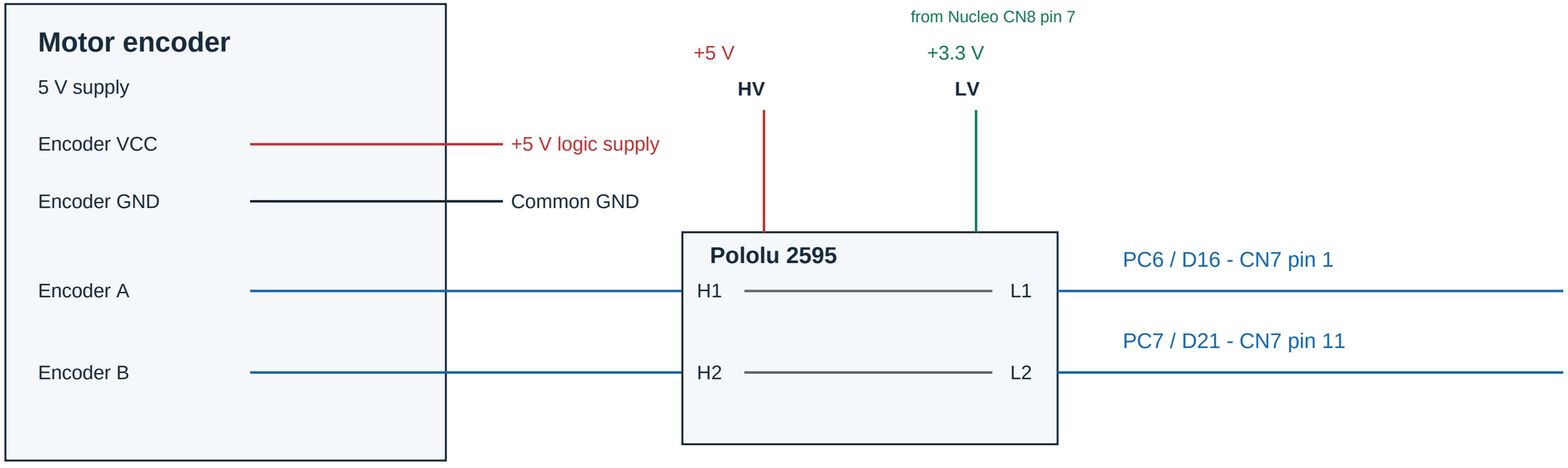
100 nF ceramic from pin 16 to GND;
100 nF ceramic from pin 8 to GND, close to IC.

GND joins: Nucleo, both supply negatives, L293B pins 4/5/12/13, and pot ground.

Driver source: ST L293B datasheet, DIP16 pinout and bidirectional motor circuit

Part 3 - Add encoder feedback

Keep the Part 2 motor-driver circuit. Remove the potentiometer. Use the onboard blue USER button (PC13).



The Pololu 2595 has no GND pin. Encoder, Nucleo, and power supplies still share common GND.

Use the module labels H1/L1 and H2/L2. Leave channels 3 and 4 unconnected.

Signal	Female socket	Header position	Connect to
PWM enable	D33 / PB0	CN10 pin 31	L293B pin 1
Motor direction 1	D10 / PD14	CN7 pin 16	L293B pin 2
Motor direction 2	D9 / PD15	CN7 pin 18	L293B pin 7
Encoder A, shifted	D16 / PC6	CN7 pin 1	Level shifter L1
Encoder B, shifted	D21 / PC7	CN7 pin 11	Level shifter L2

Before closed-loop operation: clockwise motion must increase encoder counts. Swap A/B if the sign is reversed.

Motor lead orientation sets clockwise rotation. Encoder wire colors depend on the motor cable; use its labels.

Level-shifter source: Pololu 2595 connections